

INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING
2019, ICRTAC 2019

Classification of Abnormal Masses in Ultrasonic Elastography Breast Images

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Abstract

The specificity of US (ultrasound) B-mode imaging in detecting benign masses is less due to overlap of features of malignant and benign masses. The intensity variations in a US B-mode image are due to variation in the texture of the underlying tissues, but in a US elastography image relates to strain value in tissues and have more potential in tissue differentiation. This study aims to develop a new scheme to automatically determine a seed point in an US elastography image based on mean strain value, thereby speeding up the segmentation process and to use elasticity features extracted from the segmented images to improve the classification rate. A new method of mean strain value calculation within the abnormal region is introduced to develop a new automatic method of selecting a seed point that helps in the precise segmentation of the lesion area. A neural network is used to classify the abnormal masses from texture, shape and two new elasticity features extracted from the segmented regions. The new approach of automatic seed point selection used on a dataset of 62 US elastography images achieved an accuracy of 90.32%. The average Tanimotto coefficient of US elastography images segmented using this approach is 0.75123. The classifier provided a sensitivity, specificity and classification accuracy of 80.00%, 86.67%, and 84.00% respectively. This is the first study that applies mean strain value for automatic seed point selection in US elastography images, reducing time consumed by the user and usage of elasticity features for classification of abnormal breast masses.

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Peer-review under responsibility of the scientific committee of the INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING 2019.

Keywords: Elastography, Segmentation, Seed point, Features, Classification

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1. Introduction

The specificity obtained in US B-mode imaging in detecting benign masses is less because US B-mode cannot provide much tissue differentiation. Ultrasound elastography, a new imaging technique can provide good tissue differentiation due to its strain-dependent nature. On an elastogram, harder tissues are usually displayed as darker shades of grey and more elastic or soft tissues as lighter shades of grey [1]. It is observed that both simple and complicated cysts present a bull's eye appearance with a posterior white spot [2]. A benign solid mass is soft and hence appears brighter than surrounding tissue. A malignant solid mass is harder than any of the other breast tissues and benign masses.

They appear uniformly and much darker (firmer) than the surrounding tissue and many other benign lesions [3]. Ultrasound elastography can be best applied in the evaluation of breast lesions as there is a substantial difference between fibroglandular tissue and different types of lesions [4]. The elastographic pixels in the segmented region are classified into two groups, one representing hard tissue and the other representing soft tissue [5].

A precise method of segmentation of abnormal masses of the breast is essential for accurate detection and classification of these masses. Fuzzy C-means (FCM) allows each data to be a member of every cluster but with a certain degree known as membership value [6]. Region-based level set active contour model commonly used for medical image analysis, uses statistical information of image intensity, instead of searching geometric boundaries [7].

Madabhushi et al proposed a method based on intensity and texture, for automatically selecting a seed point in US B-mode images [8]. Shan et al introduced automatic seed point selection in US B-mode images, taking into account both texture features and spatial characteristics of lesions [9]. John Massich et al described a method of selection of seed point by assigning every pixel a probability of belonging to a lesion depending on intensity, texture and geometrical constraints [10].

It is known that malignant lesions appear larger on elastograms than on B-mode images. Measuring the maximum diameter of a lesion on both the B-mode image and the corresponding elastogram allows calculation of the width ratio [11]. The two most important characteristics in real-time ultrasound elastography imaging are the size and stiffness of the lesion [12].

However, there is no method proposed for seed point detection in US elastography images. The different tissues of the breast have different stiffness and hence different modulus of elasticity and hence the strain value of tissues will be different when compressed. The grey levels on a strain image correspond to tissue elasticity. This work proposes a new scheme using the information based on mean strain value, to automatically determine a seed point inside the abnormal region of US elastography images.

The outline of the paper is as follows. Section 2 presents the workflow of a novel approach of automatic seed point selection in breast US elastography images using mean strain value. The extraction of texture, shape and elasticity features followed by classification of abnormal masses is detailed in Section 3. Finally, the performance of the proposed seed point selection, accuracy of segmentation and classification of abnormal masses is analyzed.

2. Methodology

62 US elastography and their corresponding B-mode images are obtained on a Siemens ACUSON Antares scanner with a high-frequency linear VF 7-3 transducer. This study is a retrospective analysis of stored images and has no impact on the clinical management of the patients, as all the patients have already been treated or undergone treatment. The stored images used for the study are anonymous. The images obtained from the US machine may have both normal images and images with abnormalities. It is required to select only images with abnormalities for further processing of classification.

Segregation of abnormal US B-mode images and normal images

The segregation is done automatically, by finding the difference of mean grey value between adjacent blocks in an image. The mean grey value of each 10x10 image block is determined. The difference in mean grey value between horizontally adjacent image blocks is found. The count of the number of blocks with a difference in mean

grey value >25 is found. If count is ≥ 40 , the image is classified as an image with abnormal mass; otherwise, it is classified as a normal breast image. The algorithm can correctly detect 55 images as abnormal and classifies another 7 abnormal images as normal. The 15 normal images are classified correctly. The corresponding elastography images with abnormalities are alone considered for classification into benign and malignant.

Richard Barr observed that both simple and complicated cysts present a bull's eye appearance with a posterior white spot [2]. A malignant solid mass is harder than any of the other breast tissues and benign masses. These appear uniformly and much darker (firmer) than the surrounding tissue and many other benign lesions [3]. The US elastography images are to be filtered to reduce speckle. 50 US elastography images are filtered by Lee and speckle reducing anisotropic diffusion filter (SRAD) filters respectively [13]. The mean PSNR value of Lee filtered and SRAD filtered images are 47.74 dB and 49.78 dB respectively. The SRAD filtered images are to be segmented to extract the abnormal masses in US elastography images.

Segmentation of Breast masses

FCM clustering [6] is used initially for segmentation of breast masses in US elastography images. The number of clusters is decided to be three based on subjective analysis. Of the 62 images segmented using FCM algorithm, there is no overlap between clusters in only 10 images. In all the other 52 images, the clusters overlap. Hence, this overlap in clusters decreases the intercluster variance and increases the intracluster variance in these images. Hence, it is decided to try segmenting the US elastography images with the level set method. The breast masses in 62 US elastography images are segmented using region-based level set active contour model [7]. Figures 1 and 2 show a benign solid mass segmented by both level set method and manually.

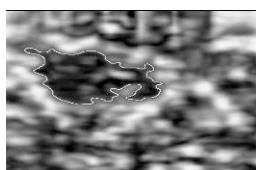


Figure 1: Results of segmenting a benign solid mass using Level set method

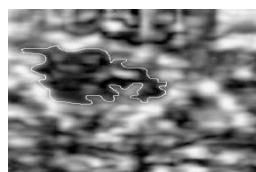


Figure 2: Results of segmenting a benign solid mass manually

The performance criterion of segmentation in US elastography images is determined using Tanimotto coefficient [14]. The value of Tanimotto coefficient of the segmented image of the benign solid mass is found to be 0.9011. Analysis of segmentation results show that for 56 images, almost correct segmentation is achieved. But, the segmentation is incomplete in 6 benign masses. This may be due to the absence of a marked separation between the normal tissue and abnormal mass in a few US elastography images as a result of desmoplastic action of masses in US elastography images [15]. The mean value of Tanimotto coefficient of such 20 region-based level set active contour segmented US elastography images is estimated to be 0.7512. However, automatic segmentation of breast lesions in 2D US elastography images done using level set active contour approach require a seed point to be selected inside the breast lesion by automatic means.

Automatic Seed point selection in US Elastography images

As the first step in the proposed new method, a seed point and an initial contour are to be detected inside an abnormal mass of US elastography image automatically to initiate level set segmentation. This requires knowledge of the elastic properties of various breast tissues and abnormal breast masses in a US elastogram as shown in Table 1. The appearance of breast tissues and abnormal masses in table 1 are about elastography strain images available from commercial US machines such as Hitachi Medical systems and Siemens [4].

Table1. Appearances of abnormal masses in a strain elastogram [3][4]

Tissue Type	Nature of tissue	Appearance in an elastogram
Simple Cyst	Central compressible material surrounded by a firm rim	Bull's eye appearance that has a bright central portion surrounded by a darker rim [4]
Complex Cyst	Firm area or area with both firm and soft regions	Uniformly dark area or as an area with both bright and dark regions [3]
Benign solid mass	Softer than surrounding tissue	Appears brighter than surrounding tissue [4]
Malignant solid mass	Stiffer than surrounding tissues and benign masses	Appears darker than surrounding tissue and benign masses [4]

Mathematical Formulation: Strain Calculation

Based on the information provided in table1, strain information in terms of pixel intensities of elastography images are computed as discussed below. To locate the seed point, as the first step, the filtered US breast elastography image of size 200 x 200 is windowed into 10 x 10 blocks. Since we have windowed into 10 x 10 blocks, the evaluation matrix becomes of size 20 x 20 and each value in the matrix gives the mean of pixel intensities of 10 x 10 pixels. Here mean is taken as an estimate of each 10 x 10 block and is obtained as per equation (1). The mean so calculated is the mean intensity value corresponding to mean strain value and hence this is denoted as mean strain value in further discussions. To obtain the approximate ranges of pixel intensity for all types of lesions in the elastography images to develop a new automatic seed selection algorithm, different threshold values have been used in Equation (1).

$$\text{Mean Strain value} = \frac{\text{sum of pixel intensities in an image block less than threshold value}}{\text{total number of pixels in an image block with intensity less than threshold value}} \quad (1)$$

The mean strain value as per Equation (1) has been computed for all the 10 x 10 blocks of US elastography images with the selected threshold values of 64, 128, 200 and 250. Tables 2, 3, 4 and 5 show the mean strain value of blocks in an US elastography image of a simple cyst, complex cyst, solid benign mass, and malignant solid mass respectively calculated with a threshold of 200. A threshold value of 200 gives a satisfactory distinction between normal breast tissue and abnormal mass.

Table 2: Mean Strain value of an elastography image of a simple cyst

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	62	42	54	127	166	147	76	67	80	78	84	67	82	95	75	48	27	35	37	112
2	109	77	106	180	180	148	76	57	62	101	127	104	103	121	106	99	70	85	64	127
3	85	96	83	149	105	71	117	166	137	108	91	69	74	65	77	77	53	67	52	55
4	91	85	54	72	57	145	183	179	129	74	55	73	82	70	70	57	69	92	133	65
5	69	75	38	33	71	179	NaN	170	86	56	70	151	135	32	42	64	119	121	192	143
6	92	79	72	30	96	198	182	172	115	57	107	191	140	53	38	66	148	102	176	165
7	79	57	47	39	90	199	176	173	175	126	123	154	106	74	87	61	85	48	71	148
8	91	54	50	59	92	190	176	182	195	187	143	112	115	150	152	113	77	68	30	105
9	67	92	140	63	62	121	130	164	199	NaN	168	159	141	141	176	151	99	58	55	84
10	47	73	110	82	76	93	79	135	159	181	183	190	179	167	177	151	147	70	50	82
11	79	67	131	165	127	180	101	43	80	140	132	154	188	NaN	NaN	186	183	83	28	63
12	119	92	182	NaN	126	181	181	133	78	94	58	112	145	169	147	154	160	64	16	45
13	147	168	160	158	119	159	190	174	113	46	71	172	135	87	45	68	127	58	17	96
14	101	138	82	68	96	57	140	183	173	133	104	178	154	95	117	137	121	50	24	97
15	41	56	125	85	50	12	59	146	NaN	180	155	187	195	174	177	182	87	52	50	52
16	34	31	85	51	78	49	52	132	178	164	129	137	168	181	145	102	39	37	25	20
17	40	34	27	49	88	80	37	45	81	86	56	82	116	139	140	101	40	17	23	32
18	28	54	40	20	11	18	41	41	35	54	62	45	55	104	112	69	18	10	25	38
19	57	51	12	8	10	13	15	23	51	35	76	37	15	21	19	11	12	20	45	36
20	54	37	31	50	64	72	45	46	31	31	38	27	28	35	41	39	20	43	69	18

This analysis is carried over after identifying the mass area in the matrix of mean strain value in comparison with the corresponding US elastography image of the breast. This comparison and analysis are carried out on 28 malignant solid masses and 34 benign masses. This analysis is carried over after identifying the mass area in the matrix of mean strain value in comparison with the corresponding US elastography image of the breast. This comparison and analysis are carried out on 28 malignant solid masses and 34 benign masses. Having estimated the desired threshold, the mean strain value in different abnormal breast masses are computed in 62 US elastography

images. The range of mean strain value in the areas of different abnormalities are analyzed and compared with background.

Table 3: Mean Strain value of an elastography image of a complex cyst

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	154	142	131	134	166	184	169	162	177	195	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2	183	174	138	123	155	190	194	193	197	NaN	NaN	NaN	199	193	187	180	177	185	NaN	NaN
3	199	193	177	166	179	196	NaN	NaN	NaN	NaN	NaN	190	176	162	148	132	119	142	180	195
4	NaN	NaN	NaN	NaN	NaN	199	197	196	192	191	189	174	134	108	95	81	69	95	147	181
5	182	188	NaN	NaN	197	187	177	174	159	145	131	95	63	51	54	59	70	82	109	142
6	141	160	189	NaN	179	152	156	162	150	131	97	73	79	97	102	97	92	84	88	120
7	119	136	176	178	159	118	120	128	132	144	134	137	160	176	176	171	144	115	104	142
8	101	114	111	96	84	90	121	136	140	149	152	176	194	NaN	NaN	196	172	142	141	170
9	96	92	78	73	61	45	56	72	83	80	95	141	182	191	NaN	198	186	159	153	174
10	109	98	72	62	58	27	16	24	37	35	27	76	158	193	NaN	NaN	194	185	177	166
11	118	127	120	114	92	28	2	23	76	108	55	33	112	180	NaN	NaN	198	195	179	163
12	104	148	179	180	137	44	9	13	22	49	33	32	120	185	188	193	197	193	192	167
13	95	145	183	185	163	137	135	136	95	54	55	83	146	156	140	150	167	159	157	143
14	42	94	146	144	135	171	197	196	180	165	137	107	111	100	83	88	101	122	130	97
15	44	80	110	99	99	155	194	NaN	NaN	198	172	111	80	61	52	62	89	118	129	76
16	64	74	76	68	82	130	169	174	188	195	174	110	77	73	91	106	130	135	121	60
17	159	145	103	73	90	116	100	96	106	109	92	72	48	82	126	140	162	144	79	19
18	154	131	108	92	86	89	48	31	41	53	56	65	60	73	102	105	103	76	23	32
19	124	114	124	139	125	115	75	35	37	45	52	66	70	70	78	78	80	72	56	94
20	89	92	129	183	182	172	143	87	75	75	73	78	69	58	67	79	81	85	103	154

Table 4: Mean Strain value of an elastography image of a benign mass

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	165	121	125	150	141	98	119	82	115	85	111	106	73	65	68	71	131	NaN	NaN	NaN
2	149	135	123	158	150	90	60	104	106	98	126	91	104	123	133	71	132	174	154	173
3	187	171	111	118	103	81	66	78	121	66	127	74	68	121	107	58	141	114	91	125
4	139	116	111	170	177	131	134	100	44	18	84	74	39	118	88	54	119	171	168	178
5	122	79	93	104	154	164	129	122	151	93	106	73	42	74	53	84	151	136	162	175
6	178	139	111	64	68	58	49	54	96	111	181	132	101	67	76	73	53	73	108	82
7	155	173	139	39	45	34	44	22	68	104	118	104	168	150	90	74	63	85	98	51
8	164	174	81	33	49	43	28	68	119	102	44	98	154	110	80	67	98	122	163	74
9	186	175	151	51	28	21	26	47	22	52	34	65	141	129	127	140	116	63	91	82
10	121	127	172	82	20	25	71	24	63	98	79	31	115	165	146	187	122	31	52	58
11	40	114	170	136	47	29	58	59	65	103	52	68	157	153	64	112	169	122	90	102
12	119	89	104	170	147	99	93	121	128	146	152	177	123	115	52	65	136	162	105	70
13	170	160	109	155	141	89	106	91	146	146	181	153	64	43	91	126	147	170	129	55
14	82	112	100	60	156	116	49	81	112	165	158	118	112	27	79	153	166	157	168	81
15	118	94	69	31	81	78	49	105	154	155	108	113	118	82	79	94	105	67	120	66
16	158	141	79	62	90	64	117	166	154	79	79	107	121	74	84	62	46	44	53	57
17	106	117	55	35	38	23	50	60	90	82	35	52	75	75	113	110	67	51	40	25
18	84	105	70	57	98	87	92	83	104	91	30	97	96	113	119	137	140	167	110	84
19	49	81	64	66	120	138	143	137	131	155	38	89	84	95	86	54	103	109	86	110
20	65	67	54	41	43	86	94	121	119	82	35	141	140	135	140	143	114	120	69	30

Table 5: Strain values of an elastography image of a malignant mass

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	48	48	43	45	53	34	32	46	44	47	43	44	105	146	175	175	162	117	132	138
2	30	34	31	17	27	9	9	26	28	11	11	20	111	161	196	195	185	144	156	164
3	68	62	35	18	13	6	6	27	26	13	13	26	118	177	NaN	196	183	139	145	163
4	80	89	28	10	11	7	7	18	24	25	16	34	119	181	NaN	191	149	97	110	144
5	78	82	30	15	8	2	5	21	18	23	21	38	97	151	187	192	138	70	78	124
6	65	72	38	12	7	3	7	8	11	23	31	49	85	109	160	182	147	83	72	112
7	68	60	26	10	14	16	17	14	15	18	24	58	92	94	118	151	142	99	84	103
8	69	47	20	15	27	29	27	23	22	12	10	41	86	116	131	161	173	148	118	105
9	60	40	21	13	16	22	22	22	18	9	5	15	59	118	157	181	198	187	168	139
10	45	35	13	2	2	6	14	14	23	41	24	6	37	114	168	196	NaN	197	193	173
11	17	20	8	1	1	1	1	2	2	29	73	52	14	24	81	141	192	178	158	175
12	2	14	30	16	1	1	1	2	29	86	64	29	29	50	109	177	171	109	140	173
13	35	88	124	103	39	9	1	2	16	54	48	30	33	60	107	154	177	99	94	159
14	114	158	166	145	109	60	12	7	9	28	57	65	79	97	101	145	183	136	136	176
15	171	190	187	165	154	122	65	47	53	73	117	147	160	154	159	177	192	183	186	196
16	188	198	196	185	181	170	132	105	92	119	168	188	191	181	189	198	198	NaN	NaN	184
17	197	182	177	189	NaN	194	179	157	153	172	194	NaN	NaN	196	198	NaN	NaN	NaN	NaN	181
18	178	138	133	175	NaN	NaN	194	188	188	193	197	191	187	179	190	NaN	NaN	NaN	NaN	169
19	154	116	109	144	181	NaN	NaN	NaN	196	177	158	148	132	146	187	198	199	NaN	NaN	162
20	101	111	111	133	159	172	176	185	182	133	96	78	70	83	130	165	165	157	159	166

The range of mean strain value thus obtained for different types of abnormal masses have been presented in table 6. The table also gives the nature of tissue in the abnormal area specified by previous researchers [3] and the level of displacement that the tissue might experience on compression. From figures 3 & 4, it is inferred that the

mean of lower and higher strain values is lowest for malignant masses due to its hardness and highest for simple cysts because of its softness.

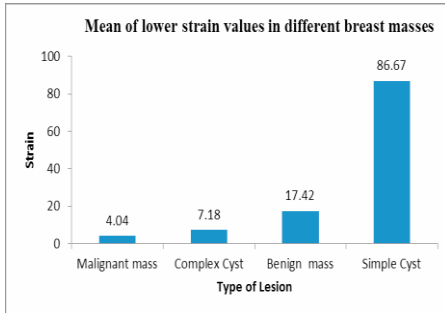


Figure 3: Mean of lower strain value in malignant solid mass, benign solid mass, simple and complex cysts

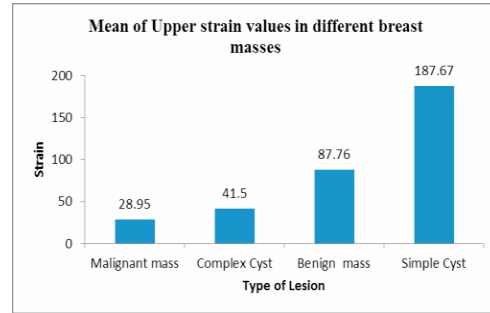


Figure 4: Mean of upper strain value in malignant solid mass, benign solid mass, simple and complex cysts

From table 6, it is observed that mean strain values obtained by the proposed new method are per the nature of abnormal masses in the breast shown in the second column.

Table 6: Range of strain values inside lesion area for different types of masses

Type of lesion	Nature of the lesion [3]	Amount of tissue displacement	Calculated mean strain value in lesion area
Malignant solid mass	Harder and less compressible than surrounding tissues	Less displacement than all other tissues	0.3 to 10
Benign solid mass	Softer and more compressible than cancer tissue	More displacement than cancer tissue	10 to 95
Simple cysts	Very soft and highly compressible	Very large displacement	140 to 190
Complex cysts	Some are hard and some are both hard and soft	less displacement or (both larger & lesser displacement)	10 to 40 (both hard and soft) 0.3 to 10 (hard)

Automatic method of Seed point selection

The automatic seed point selection proposed in the work is based on the mean strain value computed in the previous section. An US elastography image of size ($M \times N$) is filtered using SRAD to remove speckle noise and then windowed into blocks of size ($n \times n$) where $n < M, N$. 'n' is number of rows or columns in a block. A total of $(M/n \times N/n)$ blocks is obtained. In the proposed algorithm, the values of M, N , and n are taken to be 200, 200 and 10 respectively. The algorithm for the new proposed automatic seed point selection includes the following steps. The mean strain value of each 10×10 image block is calculated as per equation (1) with a threshold of 200 and thus computed for the whole image consisting of 20×20 blocks. The image blocks satisfying one of the criteria as per table 6 are considered as candidate blocks. The x and y coordinates of the candidate blocks are stored in a cell array $R\{t\}$, where 't' is the number of candidate blocks. Each of these candidate blocks is taken as the center block. A total of 8 neighboring image blocks, two on top, two at the bottom, two to the left and two to the right of the center block are considered. In the searching process, several blocks satisfying the criteria might repeat. The mode of the positions of the blocks appearing the maximum number of times is calculated to be the seed point block. The number of times each block repeats is found. The block that occurs repeatedly or the mode value is taken as the optimum seed point block to ensure that seed point block is inside the abnormal mass. A contour initialized with such a seed point helps the curve evolve to yield an accurate segmentation. The mode value of the positions of these image blocks is taken as the seed point block ($\{R2(1), R2(2)\}$) and is plotted on the US elastography image.

Initialization of Contour in Elastography Images

Accordingly, $\{R2(1), R2(2)\}$ is the seed point block selected inside the lesion area of the filtered image, by the algorithm, where $R2(2)$ and $R2(1)$ are the x and y coordinates of the seed point block. The initial contour has to

be selected around this selected seed point. The contour selected for curve evolution is a rectangle in this work and the minimum size of the initial contour that can cause curve evolution is 10 x10. Table 7 shows the formation of a square or rectangular mask, based on the location of the seed point block inside the lesion area. This serves as the initial contour. The numerical values specified are selected by analyzing several US elastography images.

Table 7: Contour initialization

Location of seed point	Position of binary mask
Centre of lesion area	$((R(2)*10)-30:(R(2)*10)+20, (R(1)*10)-20:(R(1)*10)+50))$
Near the boundary on the right inside lesion area	$((R(2)*10)-80:(R(2)*10)-30, (R(1)*10)+60:(R(1)*10)+40))$
Near the boundary on the left inside lesion area	$((R(2)*10)-30:(R(2)*10)+20, (R(1)*10)-10:(R(1)*10)+60))$
Near the boundary on the top inside lesion area	$((R(2)*10)-80:(R(2)*10)-30, (R(1)*10)+40:(R(1)*10)+90)$
Near the boundary on the bottom inside lesion area	$((R(2)*10)-50:(R(2)*10)-15, (R(1)*10)-30:(R(1)*10)+30)$

The results of seed point selection and segmentation for cyst, benign solid mass and malignant solid mass are shown in figures 5, 6, 7, 8, 9 and 10 respectively.

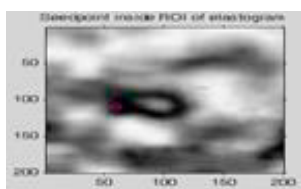


Figure 5: Seed point selection in US elastography image of a cyst

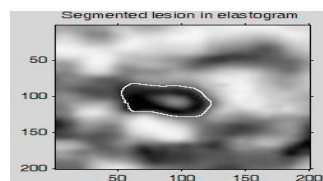


Figure 6: Segmentation in US elastography image of a cyst

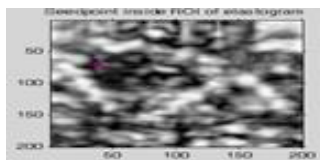


Figure 7: Seed point selection in US elastography image of a benign mass

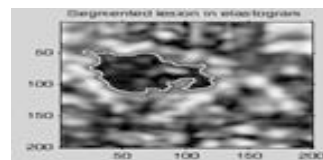


Figure 8: Segmentation in US elastography image of a benign mass

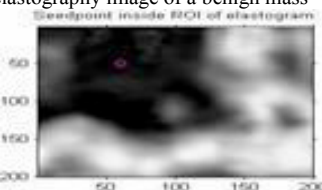


Figure 9: Seed point selection in US elastography image of a malignant mass

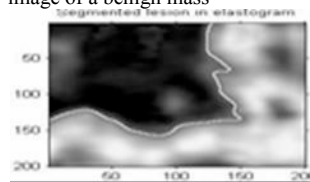


Figure 10: Segmentation in US elastography image of a malignant mass

The automatically selected seed point is shown as a small diamond-shaped structure inside the region of lesion. If the selected seed point is at the center of the mask, the algorithm segments the ROI correctly. If it is near any of the boundaries, then the user has to select that particular mask as per table 7, otherwise, the curve evolves, but segments a region outside the abnormal mass, thus leading to wrong segmentation. In a few cases, when the position of seed point detected is very closer to the boundary, the mask is to be defined in such a way that 75% of the mask is inside the region of the abnormal mass.

Thus here, user intervention is needed in selecting the initial contour appropriately from table 7, if the seed point is identified close to the boundary. Thus this proposed method reduces the time consumption on the part of the user. The approach is effective and makes segmentation of breast lesions computationally easier, more accurate and fast.

Results of Seed Point Selection Algorithm

The proposed approach of automatic seed point selection has been tested on 62 US elastography images, wherein 10 are cyst, 24 are solid benign and 28 malignant masses. The proposed approach identifies seed point correctly inside the mass area in 56 US elastography images of the breast. An accuracy of 90.32% is achieved using this approach. This technique is efficient when the different types of abnormal masses satisfy the required hardness criteria and hence the mean strain value. The average Tanimotto coefficient of US elastography images segmented using this approach is 0.75123. The results of automatic seed point selection and segmentation are shown to 4 radiologists for expert evaluation. For the evaluation procedure, scores are provided where score '5' indicates 'Excellent', '4' indicates 'Good' and '3', '2' and '1' are 'Fair', 'Bad' and 'Very Bad' respectively. An average score of 3.75 and 3.25 for automatic seed point selection and segmentation respectively is awarded.

3. Classification of Abnormal masses in breast elastogram

It is required to classify the segmented abnormal mass as benign and malignant. Therefore, after an abnormal mass being segmented from a US elastography image, features that are characteristic of both regional properties and shape are extracted from the abnormal region.

Extraction of Texture, Shape and Elasticity features

From the segmented region, GLCM texture features namely energy, entropy, dissimilarity, contrast, homogeneity & standard deviation, and grey run-length texture features namely long run emphasis, short-run emphasis, run-length nonuniformity & grey level nonuniformity are calculated. Shape features calculated are eccentricity, orientation, solidity, and width to height factor. Regional descriptors extracted from segmented regions are lesion area, perimeter, form factor, mean grey value and minimum side difference [16]. Cancers are much larger in an elastogram than on a conventional US B-mode image. This is due to the desmoplastic reaction shown by many malignant neoplasms of the breast [17]. Benign masses in an elastogram are of the same size or smaller than in US B-mode image. Based on this, a feature that shows a change in size of the lesion, namely lesion area is computed as the number of pixels belonging to an abnormal mass. Cyst is a fluid-filled mass, soft and highly compressible [4]. Soft tissues are displayed as light shades of grey (high pixel intensities) in an elastogram. A malignant solid mass is harder than any of the other abnormal masses and experiences less strain under stress and is displayed as dark shades of grey (low pixel intensities) in an elastogram. Hence the feature, mean grey value is calculated as the mean of grey level intensities inside the segmented mass.

Elasticity features namely hardness and softness are extracted from the segmented region. In an elastogram, low strain values appear as dark regions and regions with large strain appear bright [3]. The grey levels in an elastogram range from 0 to 255. In this work, it is proposed that the pixel threshold intensity of 128 is considered because pixels less than 128 are taken to be hard pixels and pixels greater than 128 are taken to be soft pixels in an US elastography image. Most of the pixels in a malignant lesion are hard pixels and hence they are dark with low grey level intensities [3]. Therefore, the mean of pixels inside the abnormal mass with intensities less than 128 is considered as hardness of the mass in the present work. This indicates that the mean of pixel intensities less than 128 in a malignant lesion is less. In an elastogram, soft tissues are displayed as light shades of grey. Hence, pixels with intensities greater than 128 are considered to be soft pixels in the present work. The mean of pixels inside the abnormal mass with intensities more than 128 is considered as softness of the mass. Most of the pixels in a benign cyst and benign mass are bright. This leads to the fact that the mean of pixel intensities greater than 128 in a benign cyst and benign mass is greater than that of a malignant lesion.

Thus, a total of 21 features are extracted from a segmented abnormal region. To select the dominant features that are statistically significant in discriminating the two classes namely benign and malignant, Student's t-test is done using SPSS version 17. This reveals that the texture features namely entropy, homogeneity, contrast and standard deviation, shape features namely lesion area and perimeter, grey level feature mean grey value, elasticity features hardness and softness are significant between the two classes. Of the 21 features extracted, nine significant features obtained by statistical analysis are used as input to a multilayer feed-forward neural network with nine input nodes, five hidden nodes and two output nodes for classifying the breast abnormalities. A learning rate of 0.05, momentum coefficient of 0.9 and a mean square error of 0.0001 is used. The neural network trained with significant features gives a sensitivity, specificity and classification accuracy of 80.00%, 86.67%, and 84.00% respectively.

4. Conclusion

Breast cancer distresses one in eight women in their course of lifetime and is in the record of common malignancy among women. Hence early detection by non-invasive comfortable methods plays a vital role. The main objective is to locate a seed point automatically in US elastography images. The proposed approach is effective and makes segmentation of breast lesions computationally easier, more accurate and fast. The main contributions are the development of a new technique for automatic seed point selection and introduction of 2 new features hardness and softness of the mass. The intensity variations in a US B-mode image are due to variation in the texture of the underlying tissues, but that is a US elastography image relates to strain value in tissues and have more potential in tissue differentiation. Hence it is decided to combine US B-mode and US elastography features for improved classification accuracy of breast masses in the next phase of work.

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